STRATA: A Process-Mining Framework for Detecting and Locating HVAC Faults from Building Trend Data

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Abstract—Faults and poor control in heating, ventilation and air-conditioning (HVAC) equipment waste 15 to 30 percent of a commercial building's energy, and many go unnoticed. Building automation systems log the operating data and display it as trend graphs, so a fault is found only when a technician reads the graphs. Automated tools either apply rules written in advance, which miss any fault that was not anticipated, or report anomaly scores that say a day is unusual without saying where or what is wrong. This paper presents STRATA, a framework that works from the trend data a building already records and learns, from one period of normal, fault-free operation, what each day should look like. It then examines every new day with eight checks, each asking whether a physical relationship or an operating sequence has moved outside its normal range. Each check is first run on fault-free days it has never seen. The number of those days it wrongly flags is its false-alarm rate. A check may then raise an alarm only when it flags far more days than that false-alarm rate could explain. Every alarm states what broke, where, the evidence and the check's measured false-alarm rate. Process mining supplies the structure. The sensor data becomes a log of events along the control sequence, and models of normal operation are discovered from that log for each zone's equipment and for the whole unit, so every alarm knows which piece of equipment it belongs to. On five public benchmark systems from Lawrence Berkeley National Laboratory, each simulated with faults whose cause and location are known, STRATA detected 158 of 175 faults, most on the first day, and raised false alarms on 26 of 480 fault-free days it had never seen. In the 50 cases where the fault lay inside one zone's equipment, the alarm pointed to the correct zone in 37, to no zone in 13, and to a wrong zone in none. Moving the framework to a new building means writing one configuration file, not new code. The planned next step is a live trial on a real building. It is also the first step toward ProcessHeal, a self-healing building that finds its own faults, locates them, works out their cause and corrects them. STRATA does the finding and the locating; working out the cause and correcting it are the later stages.

Index Terms—HVAC, fault detection, process mining, building automation, event logs, explainable alarms

I. Introduction

Buildings account for about a third of energy-related carbon emissions [1], and HVAC accounts for up to half of a building's energy use [2]. Faults and poor control that go uncorrected waste 15 to 30 percent of a commercial building's energy [3]. A study of more than 60,000 pieces of HVAC equipment found 40 percent of air handlers in a faulted state on any given day, in buildings that already had automation systems with alarms [4]. Seven of the ten most common air-handler faults in that study are faults of behaviour, such as a wrong schedule or an unreleased manual override, rather than a broken part.

A building operator today usually learns of such a fault in one of three ways. An occupant complains that a room is too hot or too cold. The operator notices that a line on a trend graph has drifted from where it usually sits. Or an automated detector raises an alarm. The first two find the fault late and depend on someone happening to look. The third works only as well as the method behind it.

Those automated methods fall into two groups. Rule libraries, which come built into commercial building automation systems, are transparent and need no training data, but they catch only the faults for which someone wrote a rule. Machine-learning detectors learn from the building's own data and so can find faults that no rule anticipated, but most report an anomaly score: a number that says today is unusual without saying where or what is wrong. A score is not enough to write a work order.

This paper presents STRATA. It is configured with rules, written once per building in a single file, but those rules do not raise alarms. They turn the sensor data into a log of events: cooling started, damper open while commanded closed, a supply-air reading that physics says cannot be right. What a normal day of those events looks like is then learned from a period of the building's own fault-free operation, and every new day is checked against it, so a fault can show up as a departure from normal even when no rule names it. Each alarm states the fault, the piece of equipment, the evidence, and how often that same check has been wrong on fault-free days.

The method behind it is process mining, which discovers a model of a process from a log of events and checks later behaviour against the model [5]. STRATA builds such a log from the sensor data, for each zone's equipment and for the whole unit. That is how an alarm knows where it belongs, and why it can check the order of control actions as well as physical relationships. To our

knowledge this is the first use of process mining for HVAC fault detection: the surveys of both fields contain no such entry, and our own search found none (Section II). On five public benchmark systems with 175 seeded faults, STRATA detected 158, most on the first day, and placed no alarm in a wrong zone (Section IV). The code and the complete evaluation are public.[1]

II. Related Work

A. Rule-based detection, the method in use today

Fault detection in real buildings is almost entirely rule-based. The air-handling-unit performance assessment rules of Schein and colleagues set the pattern [6]. A rule states a relationship that must hold in a given operating mode, such as no water flow through a closed valve, and fires when it does not. ASHRAE Guideline 36 publishes such fault conditions alongside its control sequences [7], [8], and an open-source implementation of them exists [?]. A survey of fourteen commercial fault-detection tools found this rule-and-threshold style to be the norm [9], and interviews with 29 industry experts in 2024 found it still so [10].

Detection pays when it is used. Among 26 organisations running fault-detection software on 550 buildings, the median energy saving was 8 percent [11]. Yet having the rules installed is not the same as using them. In a two-site field trial of Guideline 36, the facility team paid no attention to the built-in fault detection [12]. A 2026 California code study names alarm overload and a shortage of trained staff as the main barriers [13], and when alarms outnumber the people who can act on them, false alarms erode trust in the true ones [14].

Rules work from the first day and need no baseline data. Their limit is structural: they catch only the faults someone wrote a rule for, and miss any fault that no rule author anticipated. The Lawrence Berkeley National Laboratory (LBNL) group itself notes that commercial diagnostics often list several candidate causes or none [15], and recent work answers that with active tests that command the equipment and watch the response [16]. The same group has carried the rule approach through to correction, field-testing seven hand-coded correction algorithms on 225 pieces of equipment in four buildings [17], [18].

B. Machine learning, and what its alarms leave out

A 2024 systematic review covers 211 studies of machine learning for HVAC fault detection [2], and a 2025 review follows the field through the equipment lifecycle [19]. On the standard benchmarks, accuracy has saturated at 97 to 98 percent [20], [21]. What has not been solved is the output. A classifier or an anomaly score says that something is wrong. Where an explanation is offered, it comes from a separate layer that ranks the sensors by how much each one swayed the verdict [22], [23], [24]. That answers which signal mattered, not what the equipment should have been doing. Two 2026 studies come closest to a located alarm. One estimates the position of a stuck damper or valve after training on fault-free data alone [25]. The other weighs the evidence for each component that could be at fault and names the most likely one, without labels [26].

The second problem is labels. Real buildings have none, so the strongest recent work trains only on normal operation and flags whatever departs from it [27], [28], and what it reports is a single deviation score. A model learned on one building also travels badly to another [29], [30]: in a 2026 study, detectors that performed well inside their own buildings lost almost half of that performance on buildings they had not seen [31]. A live deployment on a Danish school building chose its detector for the fewest false alarms [32], and an audit of 65 such studies found two that shared their code [33].

Combining the two groups is not new. Two recent hybrids use hand-written rules to label historical data and then train a classifier on those labels: one on a Danish university dataset [34], the other on a real air handler in Norway with the Schein rules as its rule layer [35]. The rules are still written in advance, so the classifier can learn only the faults the rules already name, and the output is still a class label without a location. A newer direction has a large language model write and refine the rules under physical constraints and report its evidence; it beats the neural detectors on the LBNL air-handler data [36], although a 2026 review of 66 such studies found none in sustained deployment [37].

C. Process mining, and how close it has come to buildings

Process mining discovers a model of a process from a log of events and checks later behaviour against that model [5]. The discovery algorithm used here, the inductive miner, is the standard one [38]. The check lines a day's events up against the model and shows where they depart from it [39], so the explanation comes with the method rather than from a layer added afterwards. Checking conformance has served as an anomaly detector on event logs for a decade [40]. Applying it to sensor data needs one extra step, event abstraction, which turns continuous signals into discrete events [41], [42]. How to do that well, and how to keep the resulting events checkable by a person, is an open question in the process-mining community as of 2026 [43], [44].

With that step the method has reached several physical systems. It detects anomalies in chemical plants [45], in industrial alarm sequences [46] and in industrial sensor streams [47]. One group builds digital twins that learn normal behaviour offline and check it online [48], diagnoses faults on a robotic arm with few misses and few false alarms [49], and shows that features drawn from the process model make a detector far better than the plain

[1] https://github.com/iyassh/strata

check alone [50]. The closest analogue to a building is pasteurisation, where sensor signals are abstracted into activities and checked against the critical control points of the HACCP food-safety standard [51]; the parallel between those control points and Guideline 36 control sequences is the one STRATA builds on. Two kinds of work sound similar but are not this. Petri nets of HVAC equipment drawn by hand exist [52], but a net an engineer draws is not a model discovered from a log. And LBNL's OpenBuildingControl verifies that a control implementation conforms to its Guideline 36 specification before the building runs [53]; STRATA checks what the building does afterwards, from its operating data.

D. The gap, and where STRATA sits

No published work applies process mining to building HVAC fault detection. The 211-study review has no process-mining entry in its taxonomy of 23 methods [2], and the LBNL taxonomy of fault-detection methods does not list it [9]. From the other side, a review of 47 process-mining studies in Industry 4.0 [54], a review of 36 on sensor data covering industry, smart homes and healthcare [42], and a review of 93 on process management with the Internet of Things [55] contain no building application. The gap has been checked twice. Eight Google Scholar searches in February 2026 found no such paper, and a title-and-abstract search of the OpenAlex index for process mining together with HVAC, air handling, building automation, building energy or chillers, run on 16 October 2026, returns no work since 2023. The programmes of the process-mining conferences and workshops of 2024 to 2026 contain none. The nearest work analyses building-management alarm logs statistically, without discovering a process model from them [56].

The gap is not hard to explain. The two communities do not meet: fault detection sits in building science, process mining in information systems, and buildings have not been an application there. And buildings log continuous sensor streams rather than discrete events, so the event-abstraction step of Section II-C stands in the way before the method can be tried at all.

Katipamula and Brambley divide the problem into detection, isolation and identification: noticing that a fault exists, locating it, and naming its cause [3]. The rule libraries and learned detectors above stop at detection; only the two 2026 studies of Park and Paolini reach isolation. STRATA does detection and isolation, and sits between the two established groups. It learns from fault-free data what the building's normal operation is, which a rule library cannot, and it reports a location, the evidence and a measured false-alarm rate, which a score cannot. It is a complement to a rule library rather than a replacement, and its located alarms are the kind of input the LBNL correction algorithms were built to act on. Identification of the cause, and correction, are the later stages of ProcessHeal, the self-healing building toward which STRATA is the first step. Process mining has already been proposed as the basis for self-healing in the Internet of Things [57].

III. The Framework, Stage by Stage

Figure 1 shows the framework as four tiers. The top two tiers are built once per building from one fault-free period; the bottom two run on every new day. The stages below follow the figure from top to bottom.

A. Stage 1: the building's data and one configuration file

The input is the trend data a building-automation system already records, usually once a minute: temperatures, flows, pressures, damper and valve positions, fan speeds. No new sensors are needed. The framework needs one fault-free period of that data, ideally a year so that every season is seen, and one configuration file. That file is the only thing specific to a building. It maps the building's sensor names to the roles the framework understands, and it lists the control sequence of the equipment as rules, taken from the design documents, from the ASHRAE Guideline 36 sequences [7] and, as Stage 3 describes, from regularities found in the fault-free data. State rules say when the unit is occupied and which mode it is in; signature rules say which relationships must hold, such as no flow through a closed valve. Onboarding a new building means writing this file, not changing code. On the benchmarks, configuring a second equipment type took about one and a half hours, and two of the five main systems were configured before any fault data was opened, so nothing in them could have been tuned to the faults.

B. Stage 2: turning sensor data into events

Process mining works on event logs, so the framework first turns each day's minute-by-minute values into events along the control sequence, one case per day: "fan started at 06:02", "cooling became active at 09:14". State events describe what the equipment is doing, with no judgement of whether it is right. Signature events describe something that should not happen, such as water flowing through a valve commanded closed. Only state events go on to model discovery; signature events go straight to the Rules check, the dashed arrow in Fig. 1. If signature events were let in, the model would learn the written fault rules and later "detect" them, which is circular. On one system, letting them in raised the unit conformance check's apparent detection from 1.8 to 5.2 percent of fault days, and every one of the added days was the model re-finding a rule we had written ourselves.

C. Stage 3: discovering what normal looks like

From the state events of the fault-free period, the framework discovers process models of a normal day with a standard process-mining algorithm, the inductive miner [38]. It does so at two levels, the two strata that give STRATA its name: one model per terminal unit, for the normal day of that unit's damper and reheat valve, and

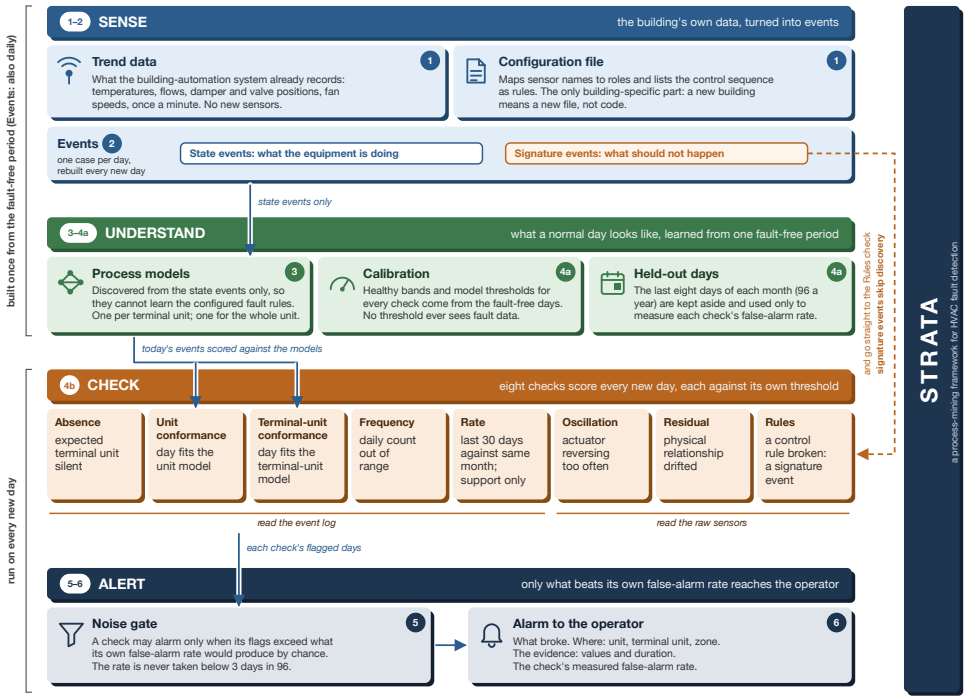


Fig. 1. The STRATA framework as four tiers. The top two are built once per building from one fault-free period; the bottom two run on every new day. The numbered badges are the stages of Section III; 4a and 4b are the two halves of Stage 4. The arrows carry the rule that matters most: state events feed model discovery; signature events never do. They skip the Understand tier and go straight to the Rules check, so a model can never re-detect a rule we wrote ourselves.

one model for the whole unit's daily cycle. Each model is a graph of the states passed through and their order, a Petri net in process-mining terms. This structure gives every alarm its location: events, rules and models are defined per terminal unit, so a check that fires on one terminal unit's data names that terminal unit. During development the models also surfaced regularities of normal operation, things always true on fault-free days, which we wrote down as rules. The models do not detect faults by themselves; Section IV-D reports the test that established this.

D. Stage 4: calibration, then eight checks

Calibration (4a). Every threshold comes from the fault-free period only. The last eight days of each month, 96 days in a year, are held out and used for nothing but measuring how often each check gives a false alarm. The remaining training days set the healthy bands and train the models. No threshold ever sees fault data, or the days on which its false-alarm rate is later reported.

The checks (4b). Each day is scored by eight checks, each named in the alarm it raises. Three read the raw sensors. Rules: a control-sequence rule was broken, such as flow through a closed valve, or a zone's airflow far from its setpoint while occupied. Residual: a physical relationship that must hold in a known mode has left its healthy band; with the cooling coil off, the supply-air temperature should equal the mixed-air temperature plus the small rise from the fan. Oscillation: an actuator reversed direction far more often than on fault-free days, the sign of an unstable control loop. Five read the event log. Unit conformance and terminal-unit conformance: how well today's events fit the discovered model; a day fits poorly when, for example, cooling starts before the fan, an order the fault-free period never showed. Absence: a terminal unit that produces events on every fault-free occupied day produced none on an occupied day. Frequency: a daily count, such as the number of heating episodes, left its healthy band. Rate: the number of the last 30 days on which a mode was active, compared with the same month of the fault-free year; it never alarms alone, because on one system it added false alarms, and is reported beside another check's alarm as supporting evidence.

E. Stage 5: the noise gate

A check may raise an alarm on a scenario only when the number of days it flagged is too large to be explained by its own false-alarm rate. The test is a binomial test, the calculation used for coin tosses. If a check wrongly flags three of 96 held-out days, flagging forty of 365 is far beyond chance and the check may alarm; flagging five is not. The rate is never taken below three in 96, so a check with no false alarms is not treated as perfect; the alarm itself reports the measured count, which can be zero, as in Figure 2. On the benchmarks the test is applied once per check per scenario, and the first day the credited check flagged is the day of the alarm. A live building has no

STRATA alarm -- series fan-powered terminal units, 2018-01-01 (day 1 of fault)
what zone primary-air flow not tracking its setpoint [check: rules]
where terminal unit TU_S, zone S
evidence 715 CFM measured against a 200 CFM setpoint, a 515 CFM excess, held for 720 min while occupied; damper 0.80 open, 0.32 on the same date of the fault-free year
rate rules check: 0 false-alarm days on the 96 held-out fault-free days

Fig. 2. One alarm from the series fan-powered benchmark, every value taken from the framework's recorded output. The seeded fault was the zone-S damper stuck at 80 percent: location and symptom correct, caught on day 1.

scenarios, only days; how the test is applied there, for example over a rolling window of recent days, is a decision for the planned trial (Section V). What is already fixed is that every check's false-alarm rate is measured on the held-out days before the first alarm and reported with each one.

F. Stage 6: the alarm the operator receives

What survives the gate reaches the operator with four things: what broke, where, the evidence, and the check's measured false-alarm rate. Figure 2 shows one, for a zone damper seeded stuck at 80 percent open. The alarm names the zone and the terminal unit, states that the primary airflow is 515 cubic feet per minute (CFM) above its setpoint and has been for twelve hours while occupied, reports the damper position of 0.80 against a normal 0.32, and says that the rules check gave zero false alarms on the 96 held-out fault-free days.

IV. Results on Public Benchmark Data

We evaluated the framework on the public HVAC fault-detection datasets from LBNL [58]. Each simulated dataset is one-minute data for one equipment type, recorded once fault-free and once per seeded fault: a stuck damper, a biased sensor, a fouled coil, a leaking valve, an unstable controller. Table I lists the seven datasets, one of them a measured year from a real rooftop unit. We found no published detection results on the fan-powered or dual-duct datasets. Unless stated, the results are for the five main systems, which have a full year of data. The simulated rooftop unit alone records the refrigerant side; the measured unit has 182 fault-free days and one fault case, so it yields a false-alarm rate but no detection count.

A. Detection and false alarms

Across the five main systems the framework detected 158 of 175 seeded faults, with 26 false-alarm days in 480 held-out fault-free days. The median time from the start of a fault to its first alarm was one day. On the real rooftop unit the false-alarm rate was 3 of 49 held-out days. A later audit of the configurations mapped every recorded column that the first configurations had left unread; sixteen of the

TABLE I
The seven LBNL datasets. "Scored" is the number of seeded faults evaluated; "Detected" those the framework caught; "False-alarm days" the held-out fault-free days on which it alarmed. †One of these was attributed on audit to a dataset defect rather than to the fault; the totals in the text count it, and are 157 without it.

System	Scored	Detected	False-alarm days
Single-duct air handler	14	14†	1 of 96
Fan-powered terminal units (parallel)	30	26	8 of 96
Fan-powered terminal units (series)	29	25	9 of 96
Dual-duct air handler	55	50	3 of 96
Fan coil unit	47	43	5 of 96
Rooftop unit (simulated)	24	20	0 of 28
Rooftop unit (measured)	1 case	—	3 of 49

158 detections came from columns that audit added. The 17 faults it missed are all coil fouling. For some coils every seeded severity was missed, because the recorded sensor values stay inside every healthy band; nothing that reads only those values could see them. On the one dataset that records the refrigerant side, the framework caught every fouling severity.

B. Where the alarm points

On the two fan-powered systems, where each air handler serves four terminal units, 50 detections made before the column audit had their seeded fault in one terminal unit. In 37 of these the alarm named the correct terminal unit, in 13 it named the air handler but no terminal unit, and in none did it name a wrong one. Over all 158 detections the rule carrying the alarm pointed at the right class of component (coil, fan, damper or terminal unit) in 83 percent of cases and at the exact seeded component in 69 percent.

C. Against two baselines

Against principal component analysis, a standard statistical method applied to the raw sensors, the framework detected 64 of 73 shared scenarios against the baseline's 61, a difference too small to be statistically significant; the baseline cannot name the terminal unit. Against the Guideline 36 fault rules that commercial systems run, implemented by an independent open-source library and scored under the same gate, the framework detected one more fault than the rules on the single-duct air handler at the same false-alarm count, and three times as many on the terminal units, because it reads the per-zone sensors the rules do not (Table II). We had predicted that STRATA would detect at least as many faults as the rules at a lower false-alarm rate on every system. It did not: on the air handler the false-alarm counts were equal, and on the terminal units the extra coverage cost eight and nine false-alarm days against the rules' none.

TABLE II
STRATA against the Guideline 36 fault rules on the three systems on which both were run, under the same gate. "Detected" is seeded faults detected; "False" is false-alarm days among the held-out days. The single-duct count is the audited 13 of Table I.

System	STRATA Detected	STRATA False	Guideline 36 rules Detected	Guideline 36 rules False
Single-duct air handler	13 of 14	1 of 96	12 of 14	1 of 96
Fan-powered terminal units (parallel)	26 of 30	8 of 96	8 of 30	0 of 96
Fan-powered terminal units (series)	25 of 29	9 of 96	9 of 29	0 of 96

D. What process mining contributes

We began with the hypothesis that the discovered process models would detect the faults themselves. They did not. Over 199 scenarios on six systems, no fault was caught by the conformance checks alone, and even on the scenario where the models did best, a one-line hand-written rule flagged more fault days than the model did. What process mining contributes is structure. The per-terminal-unit models give every alarm its location. The models surfaced regularities that became detection rules, for example that the air handler's heating never runs on an occupied workday. On the two fan-powered systems the frequency check, which counts events in the log, is one of only two checks (with the oscillation check) that see the unstable-controller faults. Process mining helps to get detection but is not detecting on its own. That finding came from pre-registered tests, with the refuting result written down before each was run.

E. Robustness to noisier data

To check that clean simulated data had not inflated the results, we applied four perturbations to the fault-free year and every fault file: sensor noise; change-of-value logging, which records a value only when it changes; gaps in the record; and shifted schedules. We then re-fitted and re-scored all five systems against a limit, set beforehand, of 10 false-alarm days in 96. Every system stayed under it, with the worst count at 7 of 96, and each condition cost at most one detection per system.

V. Deployment on a Real Building: The Next Step

This section describes planned work; none of it has been carried out. Thompson Rivers University (TRU), where the work was done, is the planned site. A 2022 energy study of the Campus Activity Centre lists an air-handling unit serving variable-air-volume terminal units, fan coil units and five rooftop units [59], all equipment types on which the framework has been evaluated. Deployment follows the tiers of Figure 1 in four steps. First, collect a fault-free period of trend data at the system's own sampling rate, a year if possible. Second, write the configuration file: every recorded point mapped to a role or excluded with a reason, and the control sequence as rules. Third, confirm silence: run the signature rules over the whole fault-free period and fix or remove any that fire, then calibrate the other checks and measure every check's false-alarm rate on this building's held-out days. Fourth, operate: each day is turned into events, scored by the eight checks and passed through the noise gate, over a window set in the trial, and what passes reaches the operator as an alarm.

Three things decide whether it works. The fault-free period must really be fault-free, because one faulty day in it can move a band edge. The configuration must read every recorded point; the column audit of Section IV showed what unread columns cost. And the silence step must run at the building's own sampling rate, because thresholds set on one-minute data do not transfer to fifteen-minute data.

VI. Limitations

Every detection result above comes from simulated data, and simulated years are cleaner than real ones; the one real building so far supports a false-alarm rate, not a detection rate. Two scoring choices were made after we saw their effect: the rate check was demoted to support-only, and the two conformance checks, whose thresholds had been set on the same days later used to measure their false-alarm rates, were re-set on separate days, which raised the unit conformance false-alarm rates on the two fan-powered systems. Slowly developing faults are caught late, up to 114 days for two fouling cases. The benchmarks seed equipment and sensor faults, not the schedule and override faults that are most common in real buildings. Finally, the alarm names the symptom and the location, not the cause; diagnosis and automated correction are the later stages of ProcessHeal.

VII. Conclusion

STRATA turns the trend data a building already records into alarms an operator can act on. It learns normal operation from one fault-free period, checks each day against physics and against discovered process models, lets through only alarms that beat a measured false-alarm rate, and states what broke, where, the evidence and that rate. On the five main benchmark systems it detected 158 of 175 seeded faults and, in the 50 cases where this could be checked, never named a wrong terminal unit. Process mining gives the framework its structure and the location on every alarm; the calibrated checks do the detecting. The next step is a live trial on a real building, planned for the TRU campus.

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References

[1] United Nations Environment Programme and Global Alliance for Buildings and Construction, "Global status report for buildings and construction 2024/2025: Not just another brick in the wall," United Nations Environment Programme, Nairobi, Tech. Rep., 2025. [Online]. Available: https://globalabc.org/resources/publications/global-status-report-buildings-and-construction-20242025-not-just-another

[2] J. Bi, H. Wang, E. Yan, C. Wang, K. Yan, L. Jiang, and B. Yang, "AI in HVAC fault detection and diagnosis: A systematic review," Energy Reviews, vol. 3, no. 2, p. 100071, 2024.

[3] S. Katipamula and M. R. Brambley, "Methods for fault detection, diagnostics, and prognostics for building systems—a review, part I," HVAC&R Research, vol. 11, no. 1, pp. 3–25, 2005.

[4] E. Crowe, Y. Chen, H. Reeve, D. Yuill, A. Ebrahimifakhar, Y. Chen, L. Troup, A. Smith, and J. Granderson, "Empirical analysis of the prevalence of HVAC faults in commercial buildings," Science and Technology for the Built Environment, vol. 29, no. 10, pp. 1027–1038, 2023.

[5] W. M. P. van der Aalst, Process Mining: Data Science in Action, 2nd ed. Springer, 2016.

[6] J. Schein, S. T. Bushby, N. S. Castro, and J. M. House, "A rule-based fault detection method for air handling units," Energy and Buildings, vol. 38, no. 12, pp. 1485–1492, 2006.

[7] ASHRAE, "ASHRAE Guideline 36-2021: High-performance sequences of operation for HVAC systems," American Society of Heating, Refrigerating and Air-Conditioning Engineers, Peachtree Corners, GA, 2021. [Online]. Available: https://www.ashrae.org/technical-resources/ashrae-standards-and-guidelines

[8] ——, "ASHRAE Guideline 36-2024: High-performance sequences of operation for HVAC systems," American Society of Heating, Refrigerating and Air-Conditioning Engineers, Peachtree Corners, GA, 2024.

[9] J. Granderson, R. Singla, E. Mayhorn, P. Ehrlich, D. Vrabie, and S. Frank, "Characterization and survey of automated fault detection and diagnostic tools," Lawrence Berkeley National Laboratory, Berkeley, CA, Tech. Rep. LBNL-2001075, Nov. 2017. [Online]. Available: https://eta-publications.lbl.gov/sites/default/files/lbnl-2001075.pdf

[10] K. Heimar Andersen, S. Pommerencke Melgaard, H. Johra, A. Marszal-Pomianowska, R. Lund Jensen, and P. Kvols Heiselberg, "Barriers and drivers for implementation of automatic fault detection and diagnosis in buildings and HVAC systems: An outlook from industry experts," Energy and Buildings, vol. 303, p. 113801, 2024.

[11] G. Lin, H. Kramer, and J. Granderson, "Building fault detection and diagnostics: Achieved savings, and methods to evaluate algorithm performance," Building and Environment, vol. 168, p. 106505, 2020.

[12] X. Zhou, C. Battisti, R. Singla, G. LaPalme, G. Paliaga, and B. Eubanks, "Field demonstration of ASHRAE Guideline 36 high-performance sequences of operation for HVAC systems," Slipstream and TRC, for the Minnesota Department of Commerce, Conservation Applied Research and Development program, Tech. Rep., 2025. [Online]. Available: https://mn.gov/commerce-stat/energy/cards-2/187281/187281_slipstream_ashrae_guideline36_final_report_ada.pdf

[13] C. Battisti, Y. Li, A. Faramarzi, R. Singla, M. Mutmansky, M. Greene, and T. Jayarathne, "2028 California Energy Code CASE report: Nonresidential HVAC fault detection and diagnostics (draft)," California Statewide Codes and Standards Enhancement Team, Tech. Rep., 2026. [Online]. Available: https://title24stakeholders.com/wp-content/uploads/2026/03/2028_CASE-Report_Nonresidential-HVAC-Fault-Detection-and-Diagnostics.pdf

[14] A. R. Diallo, L. Homri, T. Boeuf, J.-Y. Dantan, and F. Bonnet, "Quantifying and mitigating alarm fatigue caused by fault detection systems," Reliability Engineering & System Safety, vol. 267, p. 111890, 2026.

[15] G. Lin, J. House, Y. Chen, J. Granderson, and H. Zhang, "Active multi-mode data analysis to improve fault diagnosis in AHUs," Energy and Buildings, vol. 337, p. 115621, 2025.

[16] A. Parsaei, A. A. Markus, B. Gunay, W. O'Brien, R. Moromisato, and J. Bursill, "A proactive fault detection and diagnostics method for zone-level auto-commissioning in VAV AHUs," Energy and Buildings, vol. 344, p. 116027, 2025.

[17] G. Lin, M. Pritoni, Y. Chen, and J. Granderson, "Development and implementation of fault-correction algorithms in fault detection and diagnostics tools," Energies, vol. 13, no. 10, p. 2598, 2020.

[18] M. Pritoni, G. Lin, Y. Chen, R. Vitti, C. Weyandt, and J. Granderson, "From fault-detection to automated fault correction: A field study," Building and Environment, vol. 214, p. 108900, 2022.

[19] Z. Wang, Y. Qin, Y. Kong, L. Wang, Q. Leng, and C. Zhang, "Advanced fault detection, diagnosis and prognosis in HVAC systems: Lifecycle insight, key challenges, and promising approaches," Renewable and Sustainable Energy Reviews, vol. 219, p. 115867, 2025.

[20] W. Tun, J. K.-W. Wong, and S.-H. Ling, "Hybrid random forest and support vector machine modeling for HVAC fault detection and diagnosis," Sensors, vol. 21, no. 24, p. 8163, 2021.

[21] Prince, B. Yoon, and P. Kumar, "Fault detection and diagnosis in air-handling unit (AHU) using improved hybrid 1D convolutional neural network," Systems, vol. 13, no. 5, p. 330, 2025.

[22] Q. Deng, Z. Chen, W. Zhu, Z. Li, Y. Yuan, and Y. Wang, "Adaptive fusion graph convolutional network based interpretable fault diagnosis method for HVAC systems enhanced by unlabeled data," Energy and Buildings, vol. 324, p. 114901, 2024.

[23] G. Li, Z. Yao, L. Chen, T. Li, and C. Xu, "An interpretable graph convolutional neural network based fault diagnosis method for building energy systems," Building Simulation, vol. 17, no. 7, pp. 1113–1136, 2024.

[24] H. Wang, W. Yan, Y. Zhang, J. Yu, W. Gao, B. Zhou, and Y. Liu, "xMHAC-PDCNN: An explainable deep network with spatial feature embedding for fault detection and diagnosis of chillers under variable operating conditions," Building and Environment, vol. 284, p. 113504, 2025.

[25] K.-S. Park, J.-H. Kim, and C. S. Park, "Explainable fault detection and diagnosis in HVAC systems using a Koopman operator with normal data," Energy and Buildings, vol. 350, p. 116580, 2026.

[26] M. Paolini, M. S. Piscitelli, and A. Capozzoli, "A label-free hybrid fault detection and diagnosis approach for HVAC systems using Bayesian networks," Energy and Buildings, vol. 351, p. 116658, 2026.

[27] M. A. F. Abdollah, R. Scoccia, and M. Aprile, "Transformer encoder based self-supervised learning for HVAC fault detection with unlabeled data," Building and Environment, vol. 258, p. 111568, 2024.

[28] K. El Mokhtari and J. J. McArthur, "Autoencoder-based fault detection using building automation system data," Advanced Engineering Informatics, vol. 62, p. 102810, 2024.

[29] X. Zhu, K. Chen, B. Anduv, X. Jin, and Z. Du, "Transfer learning based methodology for migration and application of fault detection and diagnosis between building chillers for improving energy efficiency," Building and Environment, vol. 200, p. 107957, 2021.

[30] N. Ghalamsiah, J. Wen, K. S. Candan, T. Wu, Z. O'Neill, and A. Aghaei, "Unsupervised domain adaptation for HVAC fault diagnosis using contrastive adaptation network," Energy and Buildings, vol. 337, p. 115659, 2025.

[31] S. Jiang, "Leakage-safe cross-domain AHU fault detection for building operation: adaptation reliability under limited target information," Energy and Buildings, vol. 372, p. 118275, 2026.

[32] S. P. Melgaard, R. L. Jensen, P. Ferreira, K. H. Andersen, N. Dionísio, V. V. Cogo, and P. Heiselberg, "Fault detection in AHU: A walkthrough for implementation in a Danish educational building," Energy and Buildings, vol. 342, p. 115844, 2025.

[33] A. Mukhtar, M. Hadwiger, F. Wotawa, and G. Schweiger, "Reproducibility of machine learning-based fault detection and

diagnosis for HVAC systems in buildings: An empirical study," Energy and AI, vol. 22, p. 100658, 2025.
[34] S. Mazzetto, "Hybrid predictive maintenance for building systems: Integrating rule-based and machine learning models for fault detection using a high-resolution Danish dataset," Buildings, vol. 15, no. 4, p. 630, 2025.
[35] H. R. Y. Zabadi, F. A. Zemedkun, H. Hosamo, and D. Kraniotis, "Integration of digital twins and machine learning for predictive maintenance using APAR method rules in non-residential buildings," Frontiers in Built Environment, vol. 11, p. 1734945, 2026.
[36] S. Lin and C. Hua, "Physics-informed large language models for HVAC anomaly detection with autonomous rule generation," arXiv preprint arXiv:2510.17146, 2025. [Online]. Available: https://arxiv.org/abs/2510.17146
[37] A. Neubauer, T. Hong, H. Li, M. Yu, A. Darbandi, Y. Fürst, and M. Kriegel, "Large language models for HVAC operations in building energy systems: A critical review of methods, applications, and deployment readiness," arXiv preprint arXiv:2609.05314, 2026. [Online]. Available: https://arxiv.org/abs/2609.05314
[38] S. J. J. Leemans, D. Fahland, and W. M. P. van der Aalst, "Discovering block-structured process models from event logs — a constructive approach," in Application and Theory of Petri Nets and Concurrency (PETRI NETS 2013), ser. Lecture Notes in Computer Science, vol. 7927. Springer, 2013, pp. 311–329.
[39] J. Carmona, B. van Dongen, A. Solti, and M. Weidlich, Conformance Checking: Relating Processes and Models. Springer, 2018.
[40] F. Bezerra and J. Wainer, "Algorithms for anomaly detection of traces in logs of process aware information systems," Information Systems, vol. 38, no. 1, pp. 33–44, 2013.
[41] D. Janssen, F. Mannhardt, A. Koschmider, and S. J. van Zelst, "Process model discovery from sensor event data," in Process Mining Workshops (ICPM 2020), ser. Lecture Notes in Business Information Processing, vol. 406. Springer, 2021, pp. 69–81.
[42] E. Brzychczy, M. Aleknonytė-Resch, D. Janssen, and A. Koschmider, "Process mining on sensor data: a review of related works," Knowledge and Information Systems, vol. 67, pp. 4915–4948, 2025.
[43] T. Tratnjek, M. Kocbek Bule, and G. Polančič, "From signals to events: A maturity-based perspective on IoT-enabled process mining," Sensors, vol. 26, no. 19, p. 6293, 2026.
[44] C. Imenkamp, H. Kirchmann, S. Plumhoff, M. Weidlich, and A. Koschmider, "Glass-box activity discovery from raw sensor data," BP-Meet-IoT 2026 workshop, research track, 2026. [Online]. Available: https://bp-meet-iot.github.io/papers/bp-meet-iot2026-activity-discovery.pdf
[45] Y. Shi, N. Zhang, X. Song, H. Li, and Q. Zhu, "Novel approach for industrial process anomaly detection based on process mining," Journal of Process Control, vol. 136, p. 103165, 2024.
[46] A. Norouzifar, A. Neshastegaran, and I. Izadi, "Modeling alarm sequences using process mining techniques," in 2026 34th International Conference on Electrical Engineering (ICEE), 2026, pp. 1428–1433.
[47] E. J. Husom, A. Goknil, F. Mannhardt, S. Tverdal, S. Sen, and P. Nguyen, "Unsupervised learning and process analysis for sensor data validation in the IIoT," ACM Transactions on Internet Technology, vol. 26, no. 2, pp. 1–34, 2026.
[48] F. Vitale, S. Guarino, F. Flammini, L. Faramondi, N. Mazzocca, and R. Setola, "Process mining for digital twin development of industrial cyber-physical systems," IEEE Transactions on Industrial Informatics, vol. 21, pp. 866–875, 2025.
[49] F. Vitale, N. Dall'Ora, S. Gaiardelli, E. Fraccaroli, N. Mazzocca, and F. Fummi, "Process mining-driven modeling and simulation to enhance fault diagnosis in cyber-physical systems," Journal of Manufacturing Systems, vol. 84, pp. 189–206, 2026.
[50] F. Vitale, M. Pegoraro, W. M. P. van der Aalst, and N. Mazzocca, "Control-flow anomaly detection by process mining-based feature extraction and dimensionality reduction," Knowledge-Based Systems, vol. 310, p. 112970, 2025.
[51] A. Moradbeikie, A. P. A. d. C. Barbon, I. M. Grigore, D. F. Barbin, and S. Barbon Junior, "Process mining of sensor data for predictive process monitoring: A HACCP-guided pasteurization study case," Systems, vol. 13, no. 11, p. 935, 2025.
[52] W. Deabes, K. E. Bouazza, and W. Algthami, "Smart fuzzy Petri net-based temperature control framework for reducing building energy consumption," Sensors, vol. 23, no. 13, p. 5985, 2023.
[53] M. Wetter, P. Ehrlich, A. Gautier, M. Grahovac, P. Haves, J. Hu, A. Prakash, D. Robin, and K. Zhang, "OpenBuildingControl: Digitizing the control delivery from building energy modeling to specification, implementation and formal verification," Energy, vol. 238, p. 121501, 2022.
[54] K. Akhramovich, E. Serral, and C. Cetina, "A systematic literature review on the application of process mining to Industry 4.0," Knowledge and Information Systems, vol. 66, no. 5, pp. 2699–2746, 2024.
[55] F. De Luzi, F. Leotta, A. Marrella, and M. Mecella, "On the interplay between business process management and Internet-of-Things: A systematic literature review," Business & Information Systems Engineering, vol. 67, no. 2, pp. 265–288, 2025.
[56] Z. Khais Shahid, S. Saguna, and C. Åhlund, "Unsupervised analysis of EcoStruxure Building Operation (EBO) alarm logs in school buildings," in Proceedings of the 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '26), 2026, pp. 282–292.
[57] P. Singh, M. Saman Azari, F. Vitale, F. Flammini, N. Mazzocca, M. Caporuscio, and J. Thornadtsson, "Using log analytics and process mining to enable self-healing in the Internet of Things," Environment Systems and Decisions, vol. 42, pp. 234–250, 2022.
[58] J. Granderson, G. Lin, Y. Chen, A. Casillas, J. Wen, Z. Chen, P. Im, S. Huang, and J. Ling, "A labeled dataset for building HVAC systems operating in faulted and fault-free states," Scientific Data, vol. 10, p. 342, 2023.
[59] SES Consulting Inc., "Thompson Rivers University ASHRAE Level 1 energy study: Campus Activity Centre," Prepared for Thompson Rivers University, Vancouver, BC, Tech. Rep., May 2022. [Online]. Available: https://www.tru.ca/___shared/assets/level-1-energy-study-2022-cac64231.pdf